

COE 347 : Introduction to Computational Fluid Dynamics

OpenFoam Assignment I

Answer key

1 Non-dimensional N-S equations

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (2)$$

$$\rho u \frac{\partial v}{\partial x} + \rho v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (3)$$

Using the following non-dimensionalization

$$\tilde{x} = x/L, \tilde{y} = y/L$$

$$\tilde{u} = u/U, \tilde{v} = v/U$$

$$\tilde{p} = p/P = p/\rho U^2$$

Substituting the above in N-S equations, we get the non-dimensional form of equations as following.

$$\frac{\partial(\tilde{u}U)}{\partial(\tilde{x}L)} + \frac{\partial(\tilde{v}U)}{\partial(\tilde{y}L)} = 0 \quad (4)$$

Since U, L are constant, the non-dimensional continuity equation reads

$$\frac{\partial \tilde{u}}{\partial \tilde{x}} + \frac{\partial \tilde{v}}{\partial \tilde{y}} = 0 \quad (5)$$

Similarly for momentum equations, we get

$$\rho \tilde{u} U \frac{\partial(\tilde{u}U)}{\partial(\tilde{x}L)} + \rho \tilde{v} U \frac{\partial(\tilde{u}U)}{\partial(\tilde{y}L)} = -\frac{\partial(\tilde{p}\rho U^2)}{\partial(\tilde{x}L)} + \mu \left(\frac{\partial^2(\tilde{u}U)}{\partial(\tilde{x}L)^2} + \frac{\partial^2(\tilde{u}U)}{\partial(\tilde{y}L)^2} \right), \quad (6)$$

or after dividing through by $\rho U^2/L$,

$$\tilde{u} \frac{\partial \tilde{u}}{\partial \tilde{x}} + \tilde{v} \frac{\partial \tilde{u}}{\partial \tilde{y}} = -\frac{\partial \tilde{p}}{\partial \tilde{x}} + \frac{\mu}{\rho U L} \left(\frac{\partial^2 \tilde{u}}{\partial \tilde{x}^2} + \frac{\partial^2 \tilde{u}}{\partial \tilde{y}^2} \right) \quad (7)$$

similarly for v component

The only non-dimensional group appearing in the above equation is the Reynolds number, $Re = \rho UL/\mu$. With $Re \rightarrow 0$, the viscous term on right hand side is much larger compared to the convective terms, (Stokes flow). On the other hand, for $Re \rightarrow \infty$, the viscous term are negligible as compared to the convective terms (inviscid flow : Euler equations).

2 Description of flow for $Re = 10$

2.1 Contour plots

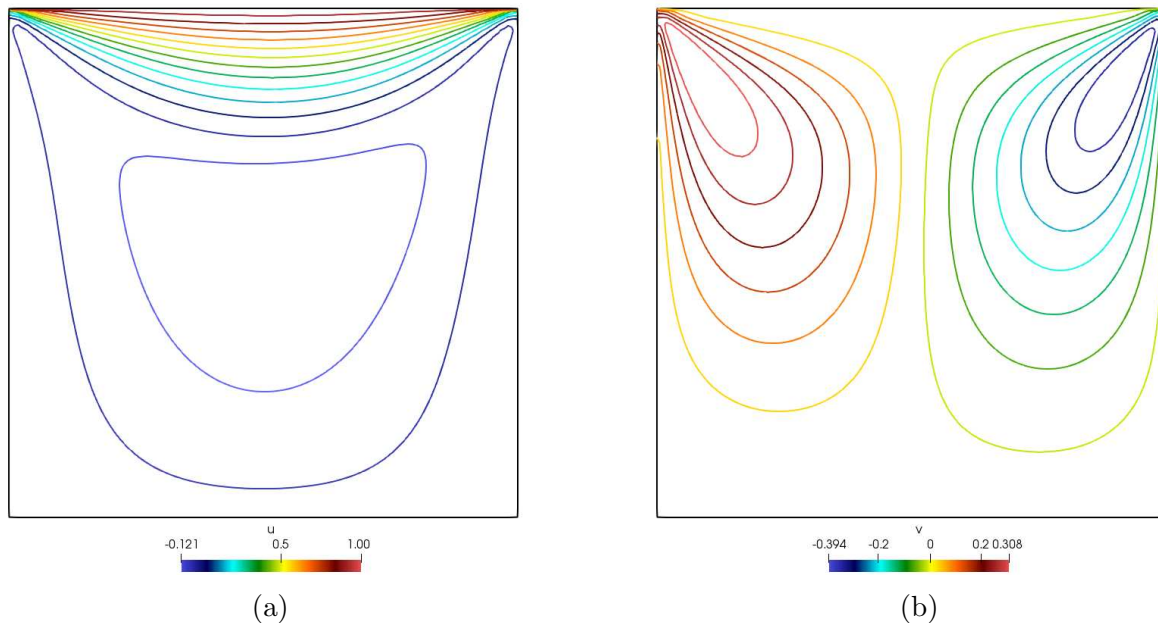


Figure 1: Contour plots for velocity components (a) $\tilde{u}$, the X component of velocity vector, and (b) $\tilde{v}$, the Y component.

Comments :

1. Apart from one or two groups, everyone plotted either the full 2D field, or streamlines. Please understand the difference between the contour lines, which are lines with constant u or v velocity, vs a streamline (where flow direction is tangent to the curve) or a 2D field plot.
2. Many of you still plotted the magnitude of $\mathbf{U}$, which is *not* same as individual components. $\mathbf{U}$ magnitude given by ParaView is $|\mathbf{U}| = \sqrt{u^2 + v^2}$, whereas you were supposed to plot contours of u, v separately.

2.2 Plots of $\tilde{u}, \tilde{v}$

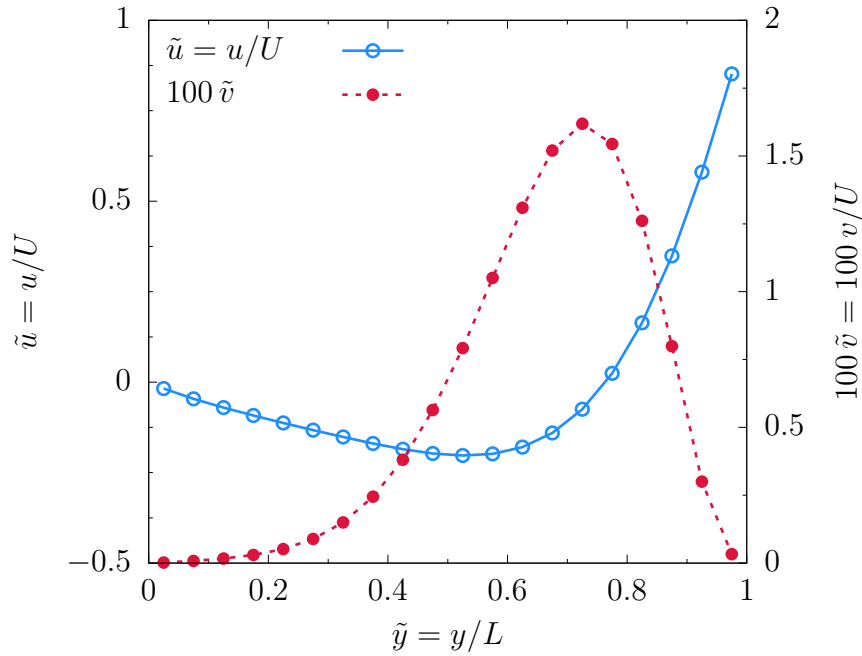


Figure 2: Non-dimensional velocity components through centerline $\tilde{x} = 0.5$ for $Re = 10$,
 $\text{---}\circ\text{---}$, $\tilde{u}$; $\text{---}\bullet\text{---}$, $\tilde{v}$.

Comments :

1. $\tilde{u}$ was almost the same for all groups, but $\tilde{v}$ was quite different. The reason behind this is that the v component near the centerline is almost zero, and thus where you evaluate $\tilde{v}$ changes the behavior from negative to positive. Note that you were supposed to plot $\tilde{v}$ at $\tilde{x} = 0.5$, which is not same as the solution at say 10^{th} or 11^{th} grid point in 20×20 grid, as the $\tilde{x} = 0.5$ plane is exactly in the middle of these two internal points. In the above plot, I averaged the solution between two nearest grid points to $\tilde{x} = 0.5$.
2. Some groups produced clear plots, but for most of the groups I could barely read the axis labels, ticks and legend text. Please make sure you select the font size in figures such that it matches the font size in main text.

3 Refining the solution

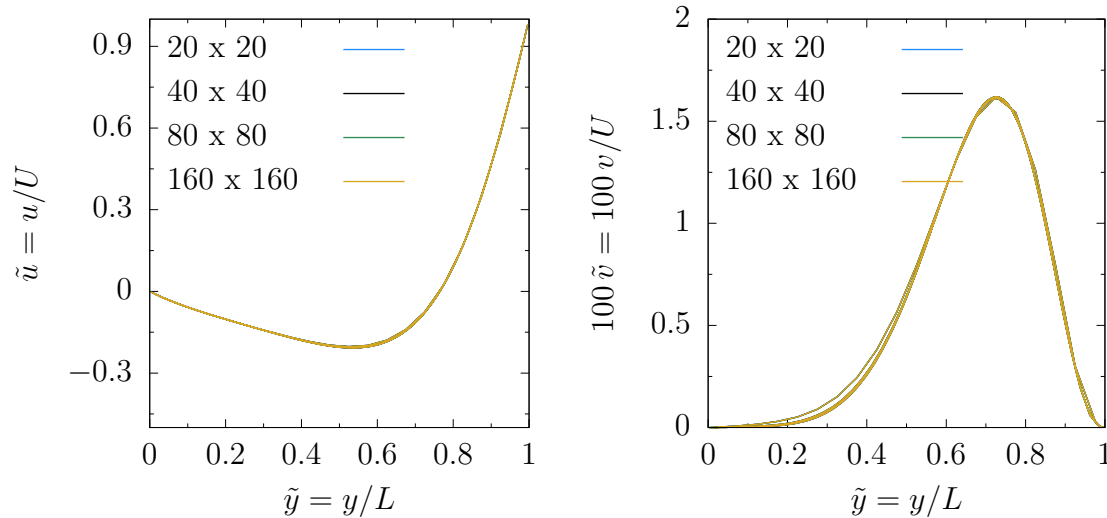


Figure 3: Non-dimensional velocity components through centerline $\tilde{x} = 0.5$ for $Re = 10$,
 $\text{---}\circ\text{---}$, $\tilde{u}$; $\text{---}\bullet\text{---}$, $\tilde{v}$.

Comments :

1. Again, plots for $\tilde{u}$ looked similar for all, but behavior of $\tilde{v}$ varied widely between the groups. Some groups found very little change in $\tilde{v}$ (which is correct), but others found it going from negative to positive, and some groups found it decreased towards zero. The latter two are because you are probably not evaluating solution at $\tilde{x} = 0.5$. Note that with finer grids, $\tilde{x}$ at the $N_x/2$ grid point moves closer to 0.5, hence **if you are plotting solution at the middle grid point it will be at different physical location with each refinement**. This is the reason some of you saw large changes in $\tilde{v}$ with grid refinement.
2. For $Re = 10$, 20×20 should be sufficient and you should not see much difference, if you did indeed plot solution at the same $\tilde{x}$ location, i.e. $\tilde{x} = 0.5$.

3.1 Computation time

Comments :

1. If equal number of steps are taken, the cost of simulation should scale as the product $N = N_x N_y = N_x^2$, as the N-S equations need to be solved on N points. However, due to stability constraints, time step needs to be appropriately reduced by the same factor of refinement to keep the CFL number under unity. Thus, the cost of integrating to

the same final actually scales as N_x^3 instead of N_x^2 (or $N^{1.5}$). This does not however, consider the fact that the finer grid simulations need on an average more iterations of the iterative solver per time step. It is hard to predict a fixed scaling law for the cost.

2. The fit numbers α, β can vary quite a bit considering performance changes from machine to machine, how frequently data is dumped, and how the code is timed itself. I recommend using the linux utility `time`, if you are running on stampede2. Also, the timing data is better if you average over multiple runs of the same simulation. I have allowed for various values of the fit quantities, due to reasons mentioned above. Almost all groups got near-full points on this, except those who did not attempt.
3. It is important to start all the simulations from same initial conditions, to get correct scaling data. Since this is a steady problem, the better your initial guess, lower the amount of iterations needed per step. Hence if you started with a mapped solution on finer mesh with `mapField` utility, you would most likely get wrong number for exponent α . Data from my runs is listed below, along with the plot.

N_x	$N = N_x N_y$	t_1 (s)	t_1 (s)	t_1 (s)	t_1 (s)	t_1 (s)	t_{av} (s)	# steps	t/step (ms)
20	400	0.228	0.274	0.235	0.318	0.269	0.265	200	1.324
40	1600	0.774	0.805	0.793	0.778	0.797	0.789	400	1.972
80	6400	5.931	5.907	5.896	6.541	6.041	6.063	800	7.578
160	25600	102.924	92.527	92.005	111.329	99.490	99.655	1600	62.284

Table 1: Timing data for different grid sizes. Each simulation is integrated from $t = 0$ to $t = 1$, and is repeated 5 times for getting time per step.

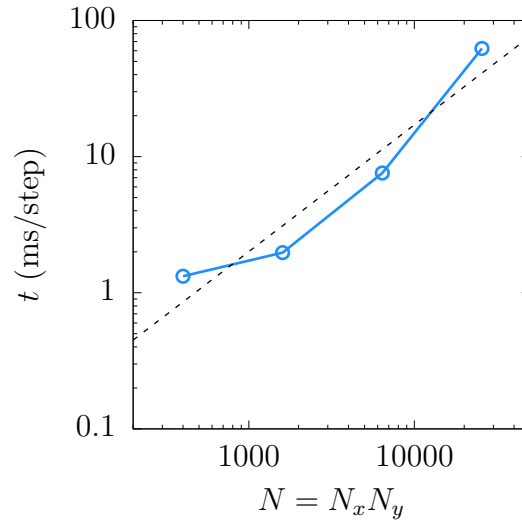


Figure 4: Timing data for $Re = 10$ case. For each grid, time per step is evaluated with 5 runs of integration from $t = 0$ to $t = 1$. For power fit, $\alpha = 0.9305$ and $\beta = 3.2 \mu s$.

4 Force on lid

Comments :

1. Many of you did not mention the numerical differentiation scheme for computing τ , or integration scheme to compute F from the shear stress. While a general idea was given to you in the problem statement itself, the choice of differentiation/integration scheme was up to you, and it was expected that you would mention which methods you used. This would have given me a better idea about the order of accuracy and would have helped me make sense of your numbers.
2. The plots I have produced here use a one sided first second order accurate method for computation of τ ,

$$\tilde{\tau}_{xy} = \frac{\partial \tilde{u}}{\partial \tilde{y}} \Big|_{\tilde{y}=1} = \frac{a + b \tilde{u}(0.5, 1 - d\tilde{y}/2) + c \tilde{u}(0.5, 1 - 3d\tilde{y}/2)}{d\tilde{y}} \quad (8)$$

where $d\tilde{y}$ is the grid spacing (normalized with $L = 0.1$). The coefficients are $a = 8/3, b = -3, c = 1/3$. This is a finite difference approximation with three points : $\tilde{y} = 1, 1 - d\tilde{y}/2$ and $1 - 3d\tilde{y}/2$.

Note that the solution field openFoam generates is *internal* field, at the cell centers of the control volumes. This is why the data does not show $\tilde{u} = 1$ at the top row of the data, since it corresponds to $\tilde{y} = 1 - d\tilde{y}/2$ and not $\tilde{y} = 1$. Notice how the solution at last point in the mid-plane line (in earlier plots) moves towards $\tilde{u} = 1$ with refinement, as $d\tilde{y}$ reduces. Maybe some of you missed this subtle issue, and that is why your shear stress and force is slightly off.

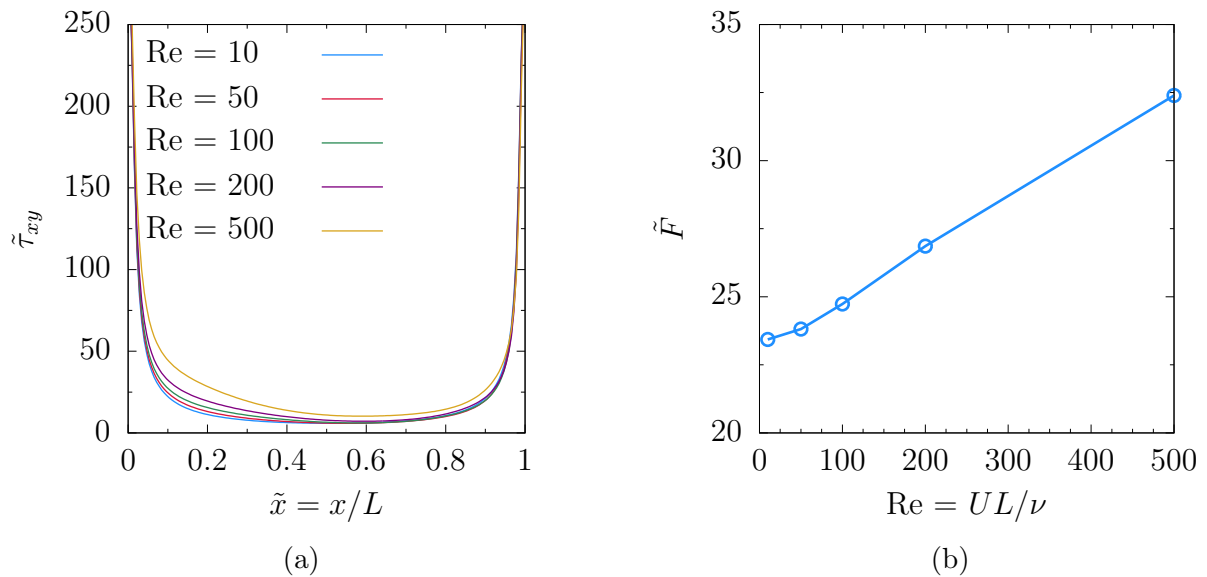


Figure 5: Non-dimentional (a) shear stress and (b) force for $Re = 10, 50, 100, 200$ and 500 .

3. It is important to match the order of accuracy in your integration and differentiation methods. If mismatched, the more accurate method will add no benefit because the error will be controlled by the lesser accurate of the two. Here I used second order finite difference method and midpoint method for integration (since τ_{xy} is computed at midpoints already), which are both second order methods. The integration simply reads

$$\tilde{F} = \sum_{i=1}^{N_x} \tau_{xy} \left(\tilde{x} = \frac{dx}{2} + (i-1)dx, \tilde{y} = 1 \right) d\tilde{x} \quad (9)$$

Note that the sum is multiplied by $\tilde{x} = 1/N_x$, and not $dx = L/N_x$.

4. The discontinuity at $\tilde{x} = 0$ and $\tilde{x} = 1$ means that the force will increase the more you refine the grid. If you ran force calculations on the courses grids, your numbers will be off as you are not adequately resolving the contributions from the edges. For the plot given here, I used 160 x 160 grid for all simulations, (although for $Re = 200$ and 500 this is not sufficient).

I was expecting clear definitions of numerical methods you used, the grid you used for these computations and so on, as these things change the solution by even a factor of two. I have taken off a few points if you simply produced these plots without any explanation of the methods.

5 Bonus question

The bonus question for 20 points was **optional**, however it indicates on canvas that the assignment was graded out of 120 points. This is just a glitch in the canvas system that is not allowing me to change the maximum points on the assignment after I have graded them. Rest assured, the assignment is weighted at 100 points. Those who score more than 100 will have the bonus points counted towards other assignments.

Depending on how you normalize the force with $H/L = 0.5, 2$, you can get different answers for $\tilde{F}$. Here I normalized with the original scale L_0 throughout, regardless of what actual width L and height H is.

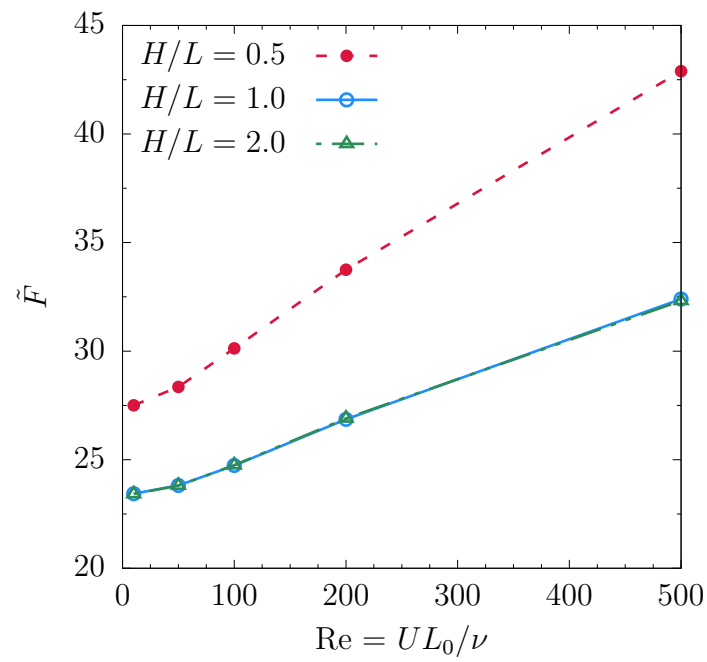


Figure 6: Non-dimensional force $\tilde{F}$ for square, shallow and tall cavity. Force changes slightly for shallow cavity, but remains the same for tall cavity.